Министерство науки и высшего образования Российской Федерации
Федеральное государственное бюджетное образовательное учреждение
высшего образования «Московский государственный технический
университет имени Н.Э. Баумана национальный исследовательский
университет» (МГТУ им. Н.Э. Баумана)

Лабораторная работа №4

Обработка телеметрической информации на примере данных альманаха GPS

Студенты группы ИУ1-41Б Соин А. Д.
Кочнева К. Д.
«16» мая 2026 г.

Преподаватель Кузнецов М. А.
«________»______________2026 г.

Москва, 2026

Содержание

1 Отчет по работе

Код написан в соответствии с заданием, строки получаются.

1.1 Проверка контрольных сумм

Ниже представлен результат чтения файла и проверки контрольных сумм.

```
 ========== CHECKSUM TEST ==========
SAT = 1 | TX = 74 | CALC = 74 | OK = 1
SAT = 2 | TX = 19 | CALC = 19 | OK = 1
SAT = 3 | TX = 66 | CALC = 66 | OK = 1
SAT = 4 | TX = 72 | CALC = 72 | OK = 1
SAT = 5 | TX = 28 | CALC = 28 | OK = 1
SAT = 6 | TX = 17 | CALC = 17 | OK = 1
SAT = 7 | TX = 66 | CALC = 66 | OK = 1
SAT = 8 | TX = 73 | CALC = 73 | OK = 1
SAT = 9 | TX = 30 | CALC = 30 | OK = 1
SAT = 10 | TX = 125 | CALC = 125 | OK = 1
SAT = 11 | TX = 46 | CALC = 46 | OK = 1
SAT = 12 | TX = 34 | CALC = 34 | OK = 1
SAT = 13 | TX = 122 | CALC = 122 | OK = 1
SAT = 14 | TX = 36 | CALC = 36 | OK = 1
SAT = 15 | TX = 114 | CALC = 114 | OK = 1
SAT = 16 | TX = 116 | CALC = 116 | OK = 1
SAT = 17 | TX = 123 | CALC = 123 | OK = 1
SAT = 18 | TX = 44 | CALC = 44 | OK = 1
SAT = 19 | TX = 34 | CALC = 34 | OK = 1
SAT = 20 | TX = 115 | CALC = 115 | OK = 1
SAT = 21 | TX = 43 | CALC = 43 | OK = 1
SAT = 22 | TX = 121 | CALC = 121 | OK = 1
SAT = 23 | TX = 114 | CALC = 114 | OK = 1
SAT = 24 | TX = 116 | CALC = 116 | OK = 1
SAT = 25 | TX = 126 | CALC = 126 | OK = 1
SAT = 26 | TX = 120 | CALC = 120 | OK = 1
SAT = 27 | TX = 115 | CALC = 115 | OK = 1
SAT = 28 | TX = 115 | CALC = 115 | OK = 1
SAT = 29 | TX = 120 | CALC = 120 | OK = 1
SAT = 30 | TX = 118 | CALC = 118 | OK = 1
SAT = 31 | TX = 46 | CALC = 46 | OK = 1
SAT = 32 | TX = 126 | CALC = 126 | OK = 1
SAT = 65 | TX = 122 | CALC = 122 | OK = 1
SAT = 66 | TX = 37 | CALC = 37 | OK = 1
SAT = 67 | TX = 119 | CALC = 119 | OK = 1
SAT = 68 | TX = 123 | CALC = 123 | OK = 1
SAT = 69 | TX = 117 | CALC = 117 | OK = 1
SAT = 70 | TX = 36 | CALC = 36 | OK = 1
SAT = 71 | TX = 119 | CALC = 119 | OK = 1
SAT = 72 | TX = 44 | CALC = 44 | OK = 1
SAT = 73 | TX = 112 | CALC = 112 | OK = 1
SAT = 74 | TX = 119 | CALC = 119 | OK = 1
SAT = 75 | TX = 120 | CALC = 120 | OK = 1
SAT = 76 | TX = 44 | CALC = 44 | OK = 1
```

```
SAT = 77 | TX = 115 | CALC = 115 | OK = 1
SAT = 78 | TX = 122 | CALC = 122 | OK = 1
SAT = 79 | TX = 36 | CALC = 36 | OK = 1
SAT = 82 | TX = 33 | CALC = 33 | OK = 1
SAT = 83 | TX = 34 | CALC = 34 | OK = 1
SAT = 84 | TX = 114 | CALC = 114 | OK = 1
SAT = 85 | TX = 118 | CALC = 118 | OK = 1
SAT = 86 | TX = 118 | CALC = 118 | OK = 1
SAT = 87 | TX = 43 | CALC = 43 | OK = 1
SAT = 88 | TX = 116 | CALC = 116 | OK = 1
SAT = 89 | TX = 119 | CALC = 119 | OK = 1
SAT = 90 | TX = 32 | CALC = 32 | OK = 1
SAT = 91 | TX = 36 | CALC = 36 | OK = 1
SAT = 92 | TX = 44 | CALC = 44 | OK = 1
```

1.2 Парсинг данных

Результаты парсинга представлены ниже, и соответствуют таблице расшифровки.

Таблица 1: Параметры GPS-альманаха

satid	week	T0a	e	Δi	$\dot{\Omega}$	$\sqrt{A}$	Ω_0	ω	M_0	af_0	af_1	health	available
1	972	5.8982e+05	0.0075502	0.027935	-7.7832e-09	0	6.0296e-05	0.017711	0.022787	2.2888e-05	2.5466e-11	false	false
2	972	5.8982e+05	0.017886	0.0070108	-7.9546e-09	2400	6.0296e-05	0.017476	0.0191	0.00017071	2.1828e-11	true	false
3	972	5.8982e+05	0.0012646	0.018354	-7.8403e-09	6976	6.0296e-05	0.021786	0.017816	1.7166e-05	2.1828e-10	true	true
4	769	2.3347e+05	0.010567	-0.0045181	-8.3203e-09	2336	6.0296e-05	0.018848	0.015575	4.1962e-05	7.276e-11	true	true
5	972	5.8982e+05	0.0052896	0.0052431	-7.9889e-09	7840	6.0296e-05	0.021703	0.022522	2.2888e-05	5.0932e-11	true	true
6	972	5.8982e+05	0.0012116	0.027672	-7.7718e-09	3616	6.0296e-05	0.017678	0.0017355	0.00019932	9.8225e-11	true	false
7	972	5.8982e+05	0.011352	0.016388	-7.8746e-09	128	6.0296e-05	0.005491	0.012862	0.00014496	1.4552e-11	true	true
8	972	5.8982e+05	0.0034409	0.026731	-7.966e-09	7648	6.0296e-05	0.013553	0.023658	0.00022507	6.5484e-11	true	false
9	972	5.8982e+05	0.0010629	0.010738	-7.9318e-09	6048	6.0296e-05	0.0012804	0.0025684	7.0572e-05	9.8225e-11	true	false
10	972	5.8982e+05	0.0034418	0.018462	-7.8175e-09	7424	6.0296e-05	0.021772	0.0069924	0.00013638	7.6398e-11	true	true
11	972	5.8982e+05	0.016735	-0.037816	-8.4689e-09	4000	6.0296e-05	0.016056	0.0075486	6.7711e-05	1.4552e-10	true	false
12	972	5.8982e+05	0.0071645	0.044521	-7.7489e-09	4736	6.0296e-05	0.0097559	0.018517	3.624e-05	2.0009e-10	true	false
13	972	5.8982e+05	0.0035424	0.026491	-7.7832e-09	6976	6.0296e-05	0.001792	0.010189	6.1989e-05	2.1464e-10	true	false
14	972	5.8982e+05	0.0095716	0.019043	-7.8632e-09	4672	6.0296e-05	0.0016295	0.0055288	0.00016785	1.4188e-10	false	true
15	972	5.8982e+05	0.010205	-0.013692	-8.1946e-09	1856	6.0296e-05	0.0010093	0.011178	2.5749e-05	2.11e-10	true	true
16	972	5.8982e+05	0.010042	0.044971	-7.7489e-09	4544	6.0296e-05	0.0098297	0.0048589	1.9073e-05	1.9645e-10	true	true
17	972	5.8982e+05	0.012275	0.040141	-7.8175e-09	7776	6.0296e-05	0.013795	0.0097728	0.00017452	6.9122e-11	false	false
18	972	5.8982e+05	0.014342	0.0062977	-7.9889e-09	7776	6.0296e-05	0.017531	0.021545	4.9591e-05	1.9281e-10	false	false
19	972	5.8982e+05	0.0098004	0.037612	-7.8289e-09	2528	6.0296e-05	0.013978	0.021109	4.3869e-05	2.0736e-10	true	true
20	972	5.8982e+05	0.0042758	-0.014411	-8.1375e-09	1152	6.0296e-05	0.021365	0.0031874	7.4387e-05	2.0736e-10	true	false
21	972	5.8982e+05	0.024491	0.0005333	-8.0232e-09	3552	6.0296e-05	0.017504	0.023496	0.00018215	9.4587e-11	false	true
22	972	5.8982e+05	0.0074401	-0.017641	-8.2061e-09	7232	6.0296e-05	0.02156	0.014341	0.00018215	2.2919e-10	false	true
23	972	5.8982e+05	0.012228	0.0015759	-8.0232e-09	512	6.0296e-05	0.0012759	0.0030331	0.00015163	1.2733e-10	false	true
24	972	5.8982e+05	0.006906	-0.00050933	-8.046e-09	6432	6.0296e-05	0.0052607	0.0018235	2.0981e-05	1.0914e-10	false	false
25	972	5.8982e+05	0.0074358	0.032166	-7.8632e-09	7744	6.0296e-05	0.0095248	0.002301	3.2425e-05	6.1846e-11	false	true
26	972	5.8982e+05	0.0027928	0.01371	-8.046e-09	5728	6.0296e-05	0.0094424	0.022085	2.861e-06	1.055e-10	true	true
27	972	5.8982e+05	0.0057969	0.034916	-7.886e-09	640	6.0296e-05	0.013584	0.01667	1.4305e-05	2.2192e-10	false	true
28	972	5.8982e+05	0.019547	0.043838	-7.7603e-09	608	6.0296e-05	0.0098465	0.023781	0.00018406	1.2005e-10	true	true
29	972	5.8982e+05	0.00065947	0.041394	-7.806e-09	2912	6.0296e-05	0.013835	0.014138	3.5286e-05	1.0186e-10	true	true

30	972	5.8982e+05	0.0032034	0.0020074	-8.0346e-09	4960	6.0296e-05	0.0056251	0.017909	0.00012207	8.7311e-11	false	false
31	972	5.8982e+05	0.0087829	0.02105	-7.8289e-09	5792	6.0296e-05	0.005539	0.011424	0.00023842	1.4552e-10	false	true
32	972	5.8982e+05	0.001924	0.015136	-7.8975e-09	6336	6.0296e-05	0.0012988	0.022697	0.00014114	7.276e-12	false	false